Counting: The Basics

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Introduction

- We have seen counting problems before
  - The size of a set: counting the elements in a set
    - e.g. the size of the Cartesian Product result
  - The weight of each digit of base-b expansion
    - e.g. number of binary integers smaller than 10000_2
  - The time complexity evaluation function: count the number of operations
    - e.g. the operations involved in a loop
  - The number of factors
    - e.g. how to get the number of factors for 648 based on its prime factorization
- We will discuss the basic rules for counting
  - The Product/Sum/Subtraction/Division Rules

The Product Rule

- A procedure can be broken down into **a sequence of two tasks**. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

Examples Applying The Product Rule$_1$

- **Example 1**: How many bit strings of length seven are there?
- **Solution**: Since each of the seven bits is either a 0 or a 1, the answer is $2^7 = 128$.
- **Example 2**: What is the number of positive binary integers smaller than 10000_2?
- **Solution**: This is the same as counting how many binary numbers can be represented by 4 digits: each digit is either 0 or 1, so the answer would be $2^4 = 16$

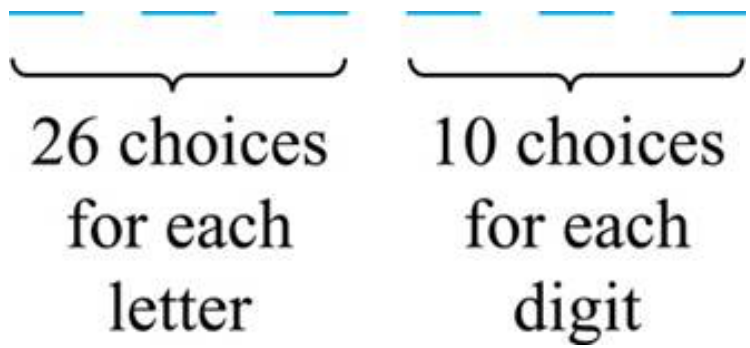
Examples Applying The Product Rule$_2$

- **Example 3: Counting One-to-One Functions**: How many one-to-one functions are there from a set with m elements to one with n elements? **Solution**: Suppose the elements in the domain are a_1, a_2,…, a_m. There are n ways to choose the value of a_1 and $n-1$ ways to choose a_2, etc. The product rule tells us that there are $n(n-1)\,(n-2)\cdots(n-m\ +1)$ such functions.

- Get a feeling how a task is divided into subproblems

Examples Applying The Product Rule$_3$

- **Example 4**: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits?
- **Solution**: By the product rule,
- there are 26 · 26 · 26 · 10 · 10 · 10 = 17,576,000 different possible license plates.



How about the plate in Texas: 4 letters + 3 digits?

Examples Applying The Product Rule$_4$

- **Example 5**: What is the value of k after the following code, where $n_1, n_2, \ldots, n_m$ are positive integers, has been executed?

```
k := 0
for i1 := 1 to n1
  for i2 := 1 to n2
    . . .
    for im := 1 to nm
      k := k + 1
```

- **Solution**: each `for` loop is a sub-task, something is repeated n_i times, and task i is to repeat task i+1 for n_i times. By product rule, the number of repetition is $n_1 n_2 \ldots n_m$, and

$k = n_1 n_2 \ldots n_m$

The Sum Rule

- If a task can be done **either in one of n_1 ways or in one of n_2 ways**, where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Examples Applying The Sum Rule$_1$

- **Example 6**: The Computer Science department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 7 members of the faculty and 403 mathematics majors and no one is both a faculty member and a student.
- **Solution**: By the sum rule it follows that there are 7 + 403 = 410 possible ways to pick a representative.

Examples Applying The Sum Rule$_2$

- **Example 7**: If $A_1, A_2, \ldots, A_m$ are pairwise disjoint finite sets, then what's the number of elements in the union of these sets?
- **Solution**: $A_1, A_2, \ldots, A_m$ are pairwise disjoint finite sets, then $A_i \cap A_j = \emptyset$ for $1 \leq i, j \leq m$. So the elements in $A_1 \cup A_2 \cup \cdots \cup A_m$ is from A_1 or A_2 or … or A_m (can't be in two sets), by applying the sum rule, we have
$$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

Combining the Sum and Product Rule$_1$

- **Example 8**: Suppose statement labels in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible labels.
- **Solution**:
  For the single letter: 26 labels.
  For the single letter + single digit: use the product rule to have $26 \cdot 10 = 260$ labels
  Either one way or another: use the sum rule
  $$26 + 26 \cdot 10 = 286$$

Combining the Sum and Product Rule$_2$

- **Example 9**: A computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. How many possible passwords are there?
- **Solution**: Let P be the total number of passwords, and let P_6, P_7, and P_8 be the passwords of length 6, 7, and 8. By the product rule:

$$P_6 = 36^6 = 2{,}176{,}782{,}336$$
$$P_7 = 36^7 = 78{,}364{,}164{,}096$$
$$P_8 = 36^8 = 2{,}821{,}109{,}907{,}456$$

By the sum rule:

$$P = P_6 + P_7 + P_8 = 2{,}901{,}650{,}853{,}924$$

The Subtraction Rule

- If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.
- This is also known as the **principle of inclusion–exclusion**

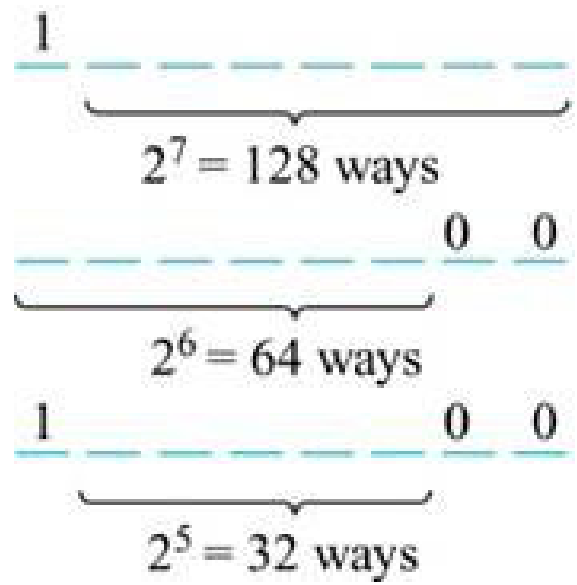
Examples of the Subtraction Rule$_1$

- **Example 10**: count the number of elements in the union of two sets A_1 and A_2.
- **Solution**: This is to count number of ways to select an element from either A_1 or A_2
  There are $|A_1|$ ways to select an element from A_1
  There are $|A_2|$ ways to select an element from A_2
  But there are some elements in both sets, so there are some common ways: $|A_1 \cap A_2|$ common elements and thus $|A_1 \cap A_2|$ common ways. So by applying the subtraction rule:

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

Examples of the Subtraction Rule$_2$

- **Example 11**: How many bit strings of length eight either start with a 1 bit or end with the two bits 00?



- **Solution**: Use the subtraction rule.
  - Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
  - Number of bit strings of length eight that end with bits 00: $2^6 = 64$
  - Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$
- Hence, the number is $128 + 64 - 32 = 160$.

The Division Rule

- There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and **for every way w, exactly d of the n ways correspond to way w**.

Examples of the Division Rule$_1$

- **Example 12**: If f is a function from A to B where A and B are finite sets, and that for every value $y \in B$ there are exactly d values $x \in A$ such that $f(x) = y$. If $|A| = n$, what's the size of B?
- **Solution**: this is *d*-to-one mapping, i.e., one value $y \in B$ can be mapped from d values $x \in A$, and by the definition of a function, we know for different value $y \in B$, they are mapped from different value $x \in A$, so by the division rule

$$|B| = |A|/d = n/d$$

Examples of the Division Rule$_2$

- **Example 13**: How many ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left and right neighbor?
- **Solution**: For table 1 there are 4 ways to assign a person, for table 2 there are 3 ways to assign another person,…. By the product rule, there are 4! = 24 ways to order the four people.
  But if we **rotate** the seating clockwise, the result is actually the same seating. For a seating, there are always the other 3 seatings the same. So by the division rule, there are 24/4 = 6 different seating arrangements (**4-to-1 mapping**).

Permutation

- **Definition**: A *permutation* of a set of distinct objects is an **ordered** arrangement of these objects. An ordered arrangement of r elements of a set is called an *r -permutation*.
- The number of r-permutations of a set with n elements is denoted by $P(n, r)$.
  - The 2-permutations of $S = \{1,2,3\}$ are $1,2$; $1,3$; $2,1$; $2,3$; $3,1$; and $3,2$. Hence, $P(3,2) = 6$.
- **Theorem 1**: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are $P(n, r) = n(n-1) \ldots (n-r+1) = n!/(n-r)!$

r-permutations of a set with n elements

Examples on Permutation$_1$

- **Example 14**: How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?
- **Solution**: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*:

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

Examples on Permutation$_2$

- **Example 15**: Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?
- **Solution**: The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

Combination$_1$

- **Definition**: An *r -combination* of elements of a set is an **unordered** selection of r elements from the set.
  - Thus, an r-combination is simply a subset of the set with r elements.
- The number of r -combinations of a set with n distinct elements is denoted by $C(n,r)$, or $\binom{n}{r}$
  - The 2-combinations of set $S = \{a,b,c,d\}$ are $\{a,b\}$, $\{a,c\}$, $\{a,d\}$, $\{b,c\}$, $\{b,d\}$, and $\{c,d\}$. Hence $C(n,r) = 6$.
  - $P(n,r) = C(n,r) \cdot P(r,r)$

Combination$_2$

- **Theorem 2**: The number of r-combinations of a set with n elements, where $n \geq r \geq 0$, equals

$$C(n,,r) = \frac{n!}{r!\,(n-r)!}$$

  - Can be proved based on the definitions of combination and permutation.
  - By the division rule

- **Corollary 1**: $C(n,r) = C(n,n-r)$
  - $n-r$ elements are the rest after picking r elements
  - This idea is called combinational proof or argument

Examples on Combination$_1$

- **Example 16**: How many poker hands of five cards can be dealt from a standard deck of 52 cards?
- **Solution**: Because the order in which the five cards are dealt from a deck of 52 cards does not matter, there are $C(52,5) = \frac{52!}{5!47!}$ different hands of give cards.

$$C(52,5) = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$
$$= 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2{,}598{,}960$$

Examples on Combination$_2$

- **Example 17**: How many bit strings of length n contain exactly r 1s?
- **Solution**: The positions of r 1s in a bit string of length n form an r -combination of the set $\{1, 2, 3, \ldots, n\}$. Hence, there are $C(n, r)$ bit strings of length n that contain exactly r 1s.

Practice 1

- 1. How many positive integers less than 1000
  - (a) are divisible by both 7 and 11?
  - (b) are divisible by exactly one of 7 and 11?

Practice 2

- 1. How many positive integers less than 1000
  - (a) are divisible by both 7 and 11?
    - Must be divisible by 77: every 77 integers, there is one such number – division rule
    - 999/77 ~= 12
      There are 12 such integers
  - (b) are divisible by exactly one of 7 and 11?
    - # divisible by 7: 999/7 ~= 142
    - # divisible by 11: 999/11 ~= 90
    - Remove those divisible by both: counted twice
      142 + 90 – 12 * 2 = 220
      There are 220 such integers

Practice 3

- 2. How many bit strings of length 10 contain either five consecutive 0s or five consecutive 1s?

Practice 4

- 2. How many bit strings of length 10 contain either five consecutive 0s or five consecutive 1s?
  - (1) five consecutive 0s can have 6 different locations (inserted between two 1s or at beginning/end)
    - At beginning: next bit must be 1, and the rest can be any 2^4=16. Similar for at the end: 16 such bit strings.
    - Between two 1s: the rest 3 bits can be any 2^3=8. There 4 such cases: 8 * 4 = 32 such bit strings.
    - There are 16 + 32 + 16 =64 such bit strings with five consecutive 0s
  - (2) similarly, there are 64 bit strings with five consecutive 1s
  - (3) 2 bit strings contains both five consecutive 0s and five consecutive 1s: 0000011111 and 1111100000
  - (4) subtraction rule to count part (3) only once:
    64 + 64 – 2 = 126
    There are 126 such bit strings

Practice 5

- 3. How many bit strings of length 10 contain
  - (a) exactly four 1s
  - (b) at most three 1s

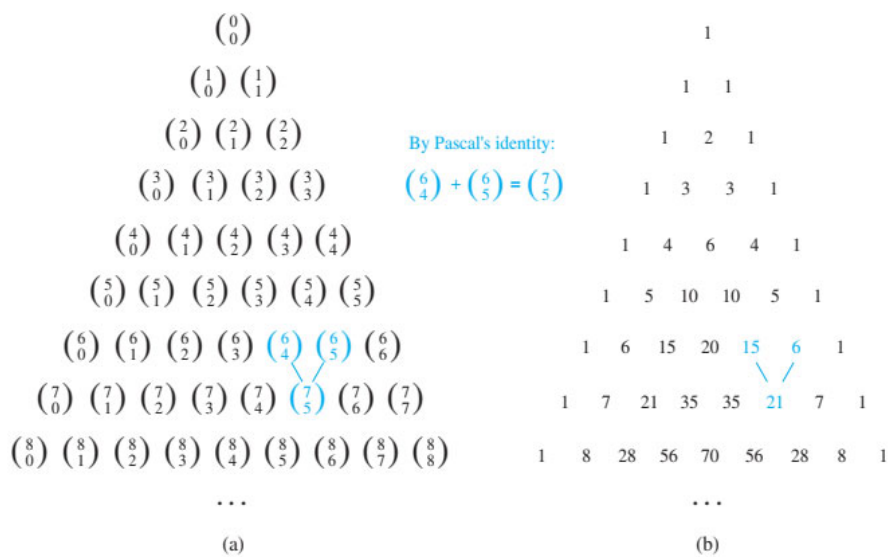
Practice 6

- 3. How many bit strings of length 10 contain
  - (a) exactly four 1s
    - We need to choose the four positions for 1s:
      C(10, 4) = 10*9*8*7/4!=210
  - (b) at most three 1s
    - Three 1s, or two 1s, or only one 1, or no 1:
      C(10, 3) + C(10,2) + C(10,1) + C(10,0)
      = 10*9*8/3! + 10*9/2! + 10 + 1
      = 120 + 45 + 10 + 1
      = 176

Extension 1

- 1. Binomial Coefficients
  - Textbook: pp 437- 440
  - $(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j$

    $= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n$
- 2. Pascal's Identity
  - Textbook: pp 440, 441
  - $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$



Extension 2

- 3. Genialized permutation and combination
  - With repetition
  - Textbook pp 445 – 450

Type	Repetition Allowed?	Formula
r-permutations	No	$P(n,r) = \frac{n!}{(n-r)!}$
r-combinations	No	$C(n,r) = \frac{n!}{r!\,(n-r)!}$
r-permutations	Yes	n^r
r-combinations	Yes	$\frac{(n+r-1)!}{r!\,(n-1)!}$

Extension 3

- 4. Coding for permutation?
  - https://www.geeksforgeeks.org/dsa/write-a-c-program-to-print-all-permutations-of-a-given-string/
  - https://leetcode.com/problems/permutations/